

VitalMoment - Project Checkpoint & Codebase

1. PROJECT OVERVIEW

VitalMoment is an offline-first AI platform for emergency first-aid.

- Perception Layer: YOLOv8 (wound detection) + MediaPipe (3D hand tracking) -> outputs Semantic JSON.
- Reasoning Layer: Medical Protocol Engine that verifies user actions via mathematical intersection.

2. ENVIRONMENT SETUP

- OS: Windows
- Python: 3.10.11 (Virtual Environment: venv)
- Dependencies: mediapipe==0.10.9, ultralytics, opencv-python

3. DIRECTORY STRUCTURE

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```
|-- data\  
|   |-- yolo_formatted\ (Roboflow dataset)  
|-- models\  
|   |-- wound_detector\weights\best.pt  
|-- scripts\  
|   |-- test_live_feed.py  
|   |-- train_yolo.py  
|-- src\  
|   |-- cv_engine\  
|       |-- anatomy_tracker.py  
|       |-- object_detector.py  
|       |-- state_builder.py  
|       |-- action_verifier.py  
|-- protocol_engine\  
    |-- engine.py
```

4. NEXT DEVELOPMENT STEPS (PHASE 2 & 3)

- Occlusion Problem: Add spatial memory so the AI remembers the wound when the hand covers it.
- Multi-Class Ontology: Add "bandage" and "gauze" to the Roboflow dataset and retrain YOLO.
- 3D Depth Checking: Use MediaPipe's Z-axis data to ensure the hand is touching the body.